24BAI70068

HOMETASK 6

(A)

hackerrank.com/challenges/placements/problem

You are given three tables: Students, Friends and Packages. Students contains two columns: ID and Name. Friends contains two columns: ID and Friend_ID (ID of the ONLY best friend). Packages contains two columns: ID and Salary (offered salary in $ thousands per month).

Column	Type
ID	Integer
Name	String

Students

Column	Type
ID	Integer
Friend_ID	Integer

Friends

Column	Type
ID	Integer
Salary	Float

Packages

Write a query to output the names of those students whose best friends got offered a higher salary than them. Names must be ordered by the salary amount offered to the

CODE:

```
SELECT s.Name

FROM Students s

JOIN Friends f ON s.ID = f.ID

JOIN Packages sp ON s.ID = sp.ID

JOIN Packages fp ON f.Friend_ID = fp.ID
```

```
WHERE fp.Salary > sp.Salary
ORDER BY fp.Salary ASC;
```

(B)

hackerrank.com/challenges/symmetric-pairs/problem

You are given a table, Functions, containing two columns: X and Y.

Column	Type
X	Integer
Y	Integer

Two pairs (X_1, Y_1) and (X_2, Y_2) are said to be symmetric pairs if $X_1 = Y_2$ and $X_2 = Y_1$.

Write a query to output all such symmetric pairs in ascending order by the value of X. List the rows such that $X_1 \leq Y_1$.

Sample Input

X	Y
20	20
20	20
20	21
23	22
22	23
21	20

CODE:

```
SELECT f1.X, f1.Y
FROM Functions f1
JOIN Functions f2 ON f1.X = f2.Y AND f1.Y = f2.X
GROUP BY f1.X, f1.Y
HAVING f1.X < f1.Y OR (f1.X = f1.Y AND COUNT(*) > 1)
ORDER BY f1.X ASC;
```